

Graphical solutions



- 1 Consider the function $y = \cos(x - 30)$.
 - a Sketch the graph of the function for $0 \leq x \leq 360$.
 - b Sketch the line $y = \frac{1}{2}$ on the same number plane.
 - c Hence, state all solutions to the equation $\cos(x - 30) = \frac{1}{2}$ over the domain $(0, 360]$.

- 2 Consider the function $y = \sin(x - 60)$.
 - a Sketch the graph of the function for $0 \leq x \leq 360$.
 - b Sketch the line $y = \frac{1}{2}$ on the same number plane.
 - c Hence, state all solutions to the equation $\sin(x - 60) = \frac{1}{2}$ over the domain $[0, 360)$.

- 3 Consider the function $y = 2 \sin 4x$.
 - a Sketch the graph of the function for $0 \leq x \leq 120$.
 - b Sketch the line $y = 1$ on the same number plane.
 - c Hence, state all solutions to the equation $2 \sin 4x = 1$ over the domain $[0, 90]$. Give your answers in degrees.

- 4 Consider the function $y = 2 \sin 2x$.
 - a Sketch the graph of the function for $0 \leq x \leq 180$.
 - b State the other function you would add to the graph in order to solve the equation $2 \sin 2x = 1$.
 - c Sketch the graph of this function on the same number plane.
 - d Hence, state all solutions to the equation $2 \sin 2x = 1$ over the domain $[-180^\circ, 180^\circ]$.

- 5 Consider the function $y = 3 \cos 2x + 1$.
 - a Sketch the graph of the function for $0 \leq x \leq 180$.
 - b State the other function you would add to the graph in order to solve the equation $3 \cos 2x + 1 = \frac{5}{2}$.
 - c Sketch the graph of this function on the same number plane.
 - d Hence, state all solutions to the equation $3 \cos 2x + 1 = \frac{5}{2}$ over the domain $[0, 180]$.

6 Consider the function $y = 2 \sin 3x - 3$.



- a Sketch the graph of the function for $0 \leq x \leq 60$.
- b State the other function you would add to the graph in order to solve the equation $2 \sin 3x - 3 = -2$.
- c Sketch the graph of this function on the same number plane.
- d Hence, state all solutions to the equation $2 \sin 3x - 3 = -2$ over the domain $[0, 60]$.

7 Consider the function $y = -2 \cos 3x$.

- a Sketch the graph of the function for $0^\circ \leq x \leq 120^\circ$.
- b State the other function you would add to the graph in order to solve the equation $-2 \cos 3x = -1$.
- c Sketch the graph of this function on the same number plane.
- d Hence, state all solutions to the equation $-2 \cos 3x = -1$ over the domain $[0^\circ, 120^\circ]$.

Exact value equations

8 State whether the following equations have a solution:

a $\cos \theta - 4 = 0$

b $9 \tan \theta + 4 = 0$

9 State the number of solutions for θ of the following equations in the domain $0^\circ < \theta < 90^\circ$.

a $\cos \theta = -\frac{1}{\sqrt{2}}$

b $\sin \theta = -\frac{\sqrt{3}}{2}$

c $\tan \theta = -1$

10 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\sin \theta = \frac{1}{\sqrt{2}}$

b $\tan \theta = \sqrt{3}$

c $\cos \theta = \frac{1}{2}$

d $\sin \theta = \frac{\sqrt{3}}{2}$

11 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:



a $\cos \theta = -\frac{1}{\sqrt{2}}$

c $\cos \theta = 0$

e $\sin \theta = 0$

g $\cos \theta = -\frac{1}{\sqrt{2}}$

i $\sin \theta = 1$

k $\tan \theta = 0$

m $4 \tan \theta + 2 = -2$

o $2 \cos \theta + 4 = 3$

b $\cos \theta = \frac{1}{2}$

d $\sin \theta = \frac{1}{2}$

f $\sin \theta = -\frac{1}{\sqrt{2}}$

h $\sin \theta = -\frac{\sqrt{3}}{2}$

j $\tan \theta = \sqrt{3}$

l $\tan \theta = -\frac{1}{\sqrt{3}}$

n $8 \cos \theta - 4 = 0$

p $8 \sin \theta - 4\sqrt{2} = 0$

Non-exact value equations

12 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\cos \theta = 0.7986$

b $\sin \theta = 0.6428$

c $\tan \theta = 0.7265$

d $\sin \theta = 0.3584$

e $\tan \theta = 2.2460$

13 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos \theta = 0.9063$

b $\cos \theta = -0.7986$

c $\sin \theta = -0.6428$

d $\sin \theta = 0.9336$

e $\tan \theta = 0.7002$

f $\tan \theta = -0.7265$